SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0615	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	M. Kalyan Ram	**Reg. No.:**	192425352
Section / Batch:	2024	**Date:**	17-7-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50

SECTION A – APPLICATION BASED QUESTIONS

Q 1. Application: Smart Traffic Management System (Detailed Explanation)

A Smart Traffic Management System uses Artificial Intelligence (AI) and sensors to monitor traffic flow, detect congestion, and optimize traffic signal timings. The system processes data collected from multiple intersections and dynamically adjusts signal patterns to reduce waiting time and improve vehicle movement.Apply the Master Theorem to solve the recurrence. Clearly identify the values of a, b, and f(n), determine the applicable case of the Master Theorem, and derive the final time complexity. Interpret the efficiency using Big-O, Big-\Theta, and Big-\Omega notations.

Program Code:

main.py

```python
def traffic_management(n):
    if n <= 1:
        return
    print(f"Processing traffic data for {n} intersections")
    traffic_management(n // 2)
    traffic_management(n // 2)
n = int(input("Enter the number of traffic intersections: "))
traffic_management(n)
print("\nTime Complexity Analysis:")
print("Recurrence Relation: T(n) = 2T(n/2) + n")
print("Using Master Theorem:")
print("a = 2, b = 2, f(n) = n")
print("Since n^(log2 2) = n, Case 2 applies.")
print("Final Complexity: Θ(n log n)")
```

Output:

Output

```
Enter the number of traffic intersections: 8
Processing traffic data for 8 intersections
Processing traffic data for 4 intersections
Processing traffic data for 2 intersections
Processing traffic data for 2 intersections
Processing traffic data for 4 intersections
Processing traffic data for 2 intersections
Processing traffic data for 2 intersections

Time Complexity Analysis:
Recurrence Relation: T(n) = 2T(n/2) + n
Using Master Theorem:
a = 2, b = 2, f(n) = n
Since n^(log2 2) = n, Case 2 applies.
Final Complexity: Θ(n log n)

=== Code Execution Successful ===
```

Q2. Application: E-Commerce Product Recommendation System

An e-commerce platform generates personalized product recommendations by analyzing customer browsing history. As the number of users increases, the recommendation engine processes user data using the recurrence:

$$T(n) = T(n-1) + n$$

Apply the **substitution method** to expand the recurrence step-by-step and derive the final time complexity. Clearly show all intermediate steps and simplify the expression. Analyze how the algorithm performs when the number of users grows rapidly. Interpret the efficiency using **asymptotic notation (Big-O, Big-Θ, and Big-Ω)** and discuss whether the recommendation system is scalable for large online shopping platforms.

Program Code:

main.py

Run

```python
def recommendation(n):
    if n == 1:
        return 1
    return recommendation(n - 1) + n
n = int(input("Enter the number of users: "))
result = recommendation(n)
print("\nTotal Processing Cost =", result)
print("\nTime Complexity Analysis:")
print("Recurrence Relation: T(n) = T(n-1) + n")
print("By Substitution Method:")
print("T(n) = 1 + 2 + 3 + ... + n")
print("T(n) = n(n+1)/2")
print("Final Complexity: Θ(n²)")
```

Output:

```
Output                                                    Clear
Enter the number of users: 5

Total Processing Cost = 15

Time Complexity Analysis:
Recurrence Relation: T(n) = T(n-1) + n
By Substitution Method:
T(n) = 1 + 2 + 3 + ... + n
T(n) = n(n+1)/2
Final Complexity: Θ(n²)

=== Code Execution Successful ===
```

Code of Conduct

I M. Kalyan Ram (192425352) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Proficient	Developing	Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						

Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Dr G Nagarajan		
Signature:	Signature:	Signature:
Date:	Date:	Date: